

Khalil LAGHA

M2 student in Electrical Energy Systems Design — power electronics & energy conversion · Université Grenoble Alpes

Lkhalilellah@gmail.com · linkedin.com/in/khalil-ellah-lagha · github.com/KhalilEllaLAGHA · khalilellahlagha.github.io · Grenoble, France ·

Authorized to work in France

PROFILE

M2 student in Electrical Energy Systems Design (UGA) specialized in **power electronics** and energy conversion applied to **EV / powertrain, electrical grids** and **AI for power systems**. Hands-on experience with DC-DC converters (buck, boost), **regulation loop tuning, efficiency optimization** and grid analysis — built across five internships, from the lab (GIPSA-lab) to industrial sites. **Seeking: work-study contract (alternance) from September 2026 — 2–3 week company / university rotation.**

EXPERIENCE

Research Intern — GIPSA-lab, Grenoble

04–07/2026 · 3.5 months

- Developed a Matlab/Simulink model of a PEMFC fuel cell — **6 %** dynamic / **1 %** static error.
- Tuned the regulation loop of a PI-controlled **DC-DC boost converter** for voltage regulation, targeting conversion-efficiency optimization — validated in simulation, then on real hardware.

Intern — SLB / Schlumberger

07/2025 · 1 month

- Completed an international industrial immersion; authored technical summaries (directional drilling); earned NEST & SIPP Level 2 HSE certifications.

Power Electronics & Automation Intern — EURL Lagha

01/2024 · 15 days + 09/2024 · 1 month

- Contributed to the design of an **industrial DC-DC buck converter** — 50 V → 0–48 V adjustable, **25 A**.
- Worked on **4 transformer-rectifiers** feeding pipelines (electrolysis): 400 V three-phase → 100 V, rectification + filtering (ripple < 5 %, EMI compliance) — **100 V DC / 80 A (≈ 8 kW)** output.
- Co-designed the supervision HMI in Siemens TIA Portal; produced the electrical schematics in EPLAN.

Power Grid Intern — Sonelgaz

03/2024 · 15 days · observation

- Studied substation architecture and the full generation → distribution chain (**400/220 kV → 60 kV → 30/10 kV → 400/230 V**); visited an MV/LV substation.

EDUCATION

M2 CSEE — Electrical Energy Systems Design · Université Grenoble Alpes

09/2026 · admitted

M1 EEA (EECS) · Université Grenoble Alpes

2025 – 2026

Ranked 2nd of the cohort · annual average **15.43/20**.

Engineering programme in Electrical Engineering · École Nationale Polytechnique, Algiers

2021 – 2025

4 of 5 years completed · moved to France for the master's · prep classes: among the top-ranked.

Baccalauréat in Mathematics

2021

Highest honours (Très Bien) — **16.93/20**.

PROJECTS

Power grid analysis — Matlab · solo **FLAGSHIP**

- Performed load-flow analysis from **3 to 118 buses** (PQ/PV/slack); compared **3 numerical methods** (accuracy vs computation time); ran transient-stability studies.

Field-oriented control (FOC) of an induction motor — Matlab/Simulink

- Implemented Park transforms, PID controllers and observers; tuned the cascaded current- and speed-regulation loops.

Electrical machine sizing — Python · team

- Designed the stator winding; computed flux density via reluctance networks; automated parameter generation.

Also: tuned the cascade control of a DC motor — three-phase thyristor bridge, outer speed loop / inner current loop (Simulink); built a Siemens Step7 automation project (HMI + Ladder, simulation) and an Arduino line-follower robot.

SKILLS

Power conversion (all types) · power conversion for the electrical grid · power electronics for machine control · electrical machines & transformers · power system analysis, transient stability · control engineering — regulation loop tuning · industrial automation (PLC, HMI/SCADA) · AI (neural networks, genetic algorithms, fuzzy logic)

SOFTWARE

Matlab/Simulink · LTspice · Proteus · Python · EPLAN · AutoCAD Electrical · Siemens TIA Portal · Step7 (Ladder/SCL) · SolidWorks · Arduino · C/C++

SOFT SKILLS & ACTIVITIES

Fast learner · comfortable with new tools & software · problem solving · rigorous · organized — IEEE Student Branch ENP · VIC (ENP)

LANGUAGES & CERTIFICATIONS

French — C1 · English — C2 · Arabic — native | SLB NEST & SIPP Level 2 — 07/2025 (HSE)